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Redes de Internet / Internet Networks
Practical Lesson No. 9: BGP and OSPF

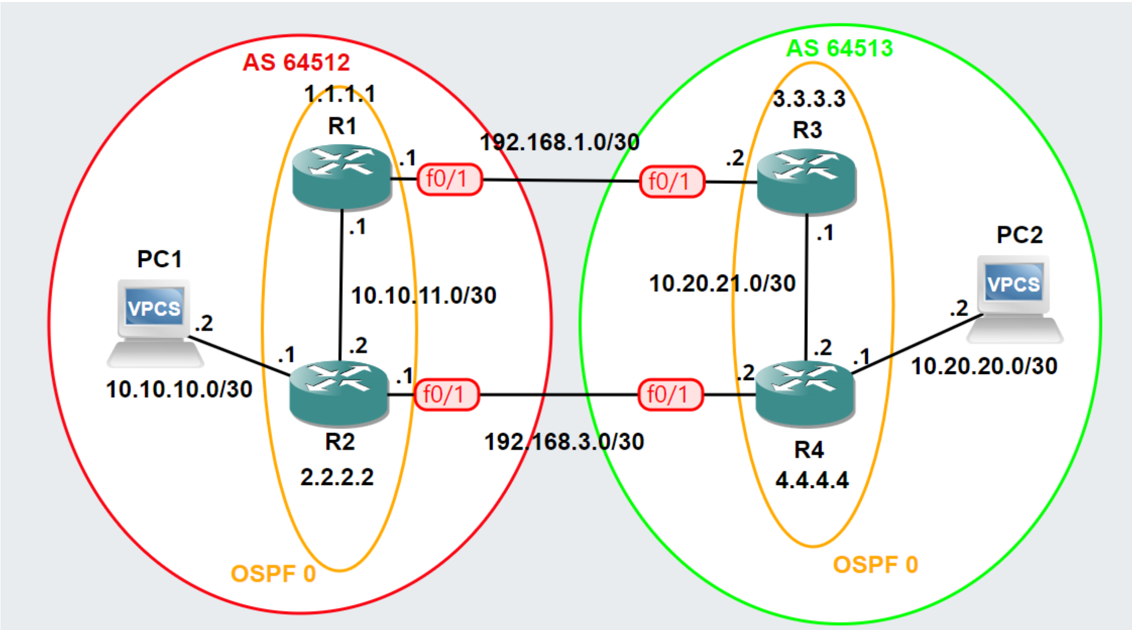
Context

In this activity, we will use the GNS3 simulator to test some of the features of BGP (Border Gateway Protocol) based on a set of routes learned via iBGP or eBGP. The former refers to neighbor relationships within an autonomous system (AS), while the latter refers to the relationship between different autonomous systems on the Internet.

Given that iBGP requires the existence of a “full mesh” of connections in the local routing system, or alternatively the use of mitigation options such as “route reflectors,” it is common to use an Interior Gateway Protocol (IGP), such as OSPF, as the iBGP protocol. This solution not only converges much faster than BGP but also has the advantage of allowing neighborhoods to be defined on loopback interfaces, which never go down, creating multiple possible paths for routing.

1 – Basic Topology

The base topology is available in the script “BGP_Aula2.gns3project” via the link. This topology should be imported using the “import portable project” option and includes two autonomous systems (AS) that will communicate via eBGP. Within each AS, an OSPF area will be configured using process number 1. All routers have their loopback interfaces configured, as shown in the figure.



2 – OSPF configuration for iBGP

First, we will configure the OSPF protocol to support internal iBGP distribution. To do this, we will use the familiar router ospf <proc id> command. As mentioned in the box, Loopback interfaces will be used in the OSPF configuration as iBGP, given their properties of never going down. These interfaces will be configured as passive using the passive-interface Loopback command to ensure that these interfaces stop sending OSPF hello packets and do not establish new OSPF adjacencies with the neighboring autonomous system, although they announce the routes with which they are interconnected. The log-neighbor-changes command is also used to enable/disable logging messages that are generated when a neighbor's status changes. The internal interfaces where OSPF hello packets will be allowed will be configured with the inverted mask corresponding to the /30 subnet (255.255.255.252) -> 0.0.0.3.

R1 Configuration

```
R1 (config)# router ospf 1
R1 (config-router)# log-adjacency-changes
R1 (config-router)# passive-interface Loopback0
R1 (config-router)# network 1.1.1.1 0.0.0.0 area 0
R1 (config-router)# network 10.10.11.0 0.0.0.3 area 0
```

R2 Configuration

```
R2 (config)# router ospf 1
R2 (config-router)# log-adjacency-changes
R2 (config-router)# passive-interface Loopback0
R2 (config-router)# network 2.2.2.2 0.0.0.0 area 0
R2 (config-router)# network 10.10.11.0 0.0.0.3 area 0
```

R3 Configuration

```
R3 (config)# router ospf 1
R3 (config-router)# log-adjacency-changes
R3 (config-router)# passive-interface Loopback0
R3 (config-router)# network 3.3.3.3 0.0.0.0 area 0
R3 (config-router)# network 10.20.21.0 0.0.0.3 area 0
```

R4 Configuration

```
R4 (config)# router ospf 1
R4 (config-router)# log-adjacency-changes
R4 (config-router)# passive-interface Loopback0
R4 (config-router)# network 4.4.4.4 0.0.0.0 area 0
R4 (config-router)# network 10.20.21.0 0.0.0.3 area 0
```

3 – Checking the OSPF configuration

Use the show ip ospf database and show ip interface brief commands to view the settings you have made. You can optionally use debug ip ospf adj command to view the OSPF messages exchanged.

Example:

```
R2#sh ip ospf database

            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1452        0x80000005 0x00A226 2
2.2.2.2         2.2.2.2         1463        0x80000005 0x00B605 2

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.10.11.1      1.1.1.1         1452        0x80000003 0x007397
```

4 – BGP peering configuration (iBGP + eBGP)

BGP configuration begins with the router bgp [ASN] command. Next, the bgp log-neighbor-changes command is used to enable/disable logging messages that are generated when the status of a BGP neighbor changes. The no synchronization command is used to indicate that iBGP routers do not need to wait for synchronization with the internal routing protocol, such as OSPF, since there is no traffic passing through a transit AS.

The BGP neighbor relationship is defined using the neighbor remote-as command, which initiates the peering connection. The update-source Loopback command forces BGP peering connections with the neighbor to use the IP address of the loopback interface, thus benefiting from the inherent resilience of these interfaces.

In eBGP sessions, associations are made using the IP addresses of the interfaces directly connected between the routers. The next-hop-self-command is particularly important in iBGP relationships, even within the same AS, as it ensures that the next hop for the advertised routes is the router itself. This avoids problems where the neighboring router may not have a route to the original announced next hop, ensuring that traffic is correctly routed through the IGP, such as OSPF.

Finally, the no auto-summary command is used to ensure that routes advertised via BGP are never summarized, which is recommended to avoid problems in networks with discontinuous addressing.

R1 Configuration

```
R1 (config)# router bgp 64512
R1 (config-router)# no synchronization
R1 (config-router)# bgp log-neighbor-changes
R1 (config-router)# neighbor 2.2.2.2 remote-as 64512
R1 (config-router)# neighbor 2.2.2.2 update-source Loopback0
R1 (config-router)# neighbor 2.2.2.2 next-hop-self
R1 (config-router)# neighbor 192.168.1.2 remote-as 64513
R1 (config-router)# no auto-summary
```

R2 Configuration

```
R2 (config)# router bgp 64512
R2 (config-router)# no synchronization
R2 (config-router)# bgp log-neighbor-changes
R2 (config-router)# neighbor 1.1.1.1 remote-as 64512
R2 (config-router)# neighbor 1.1.1.1 update-source Loopback0
R2 (config-router)# neighbor 1.1.1.1 next-hop-self
R2 (config-router)# neighbor 192.168.3.2 remote-as 64513
R2 (config-router)# no auto-summary
```

R3 Configuration

```
R3 (config)# router bgp 64513
R3 (config-router)# no synchronization
R3 (config-router)# bgp log-neighbor-changes
R3 (config-router)# neighbor 4.4.4.4 remote-as 64513
R3 (config-router)# neighbor 4.4.4.4 update-source Loopback0
R3 (config-router)# neighbor 4.4.4.4 next-hop-self
R3 (config-router)# neighbor 192.168.1.1 remote-as 64512
R3 (config-router)# no auto-summary
```

R3 Configuration

```
R4 (config)# router bgp 64513
R4 (config-router)# no synchronization
R4 (config-router)# bgp log-neighbor-changes
R4 (config-router)# neighbor 3.3.3.3 remote-as 64513
R4 (config-router)# neighbor 3.3.3.3 update-source Loopback0
R4 (config-router)# neighbor 3.3.3.3 next-hop-self
R4 (config-router)# neighbor 192.168.3.1 remote-as 64512
R4 (config-router)# no auto-summary
```

5 – Checking the BGP configuration

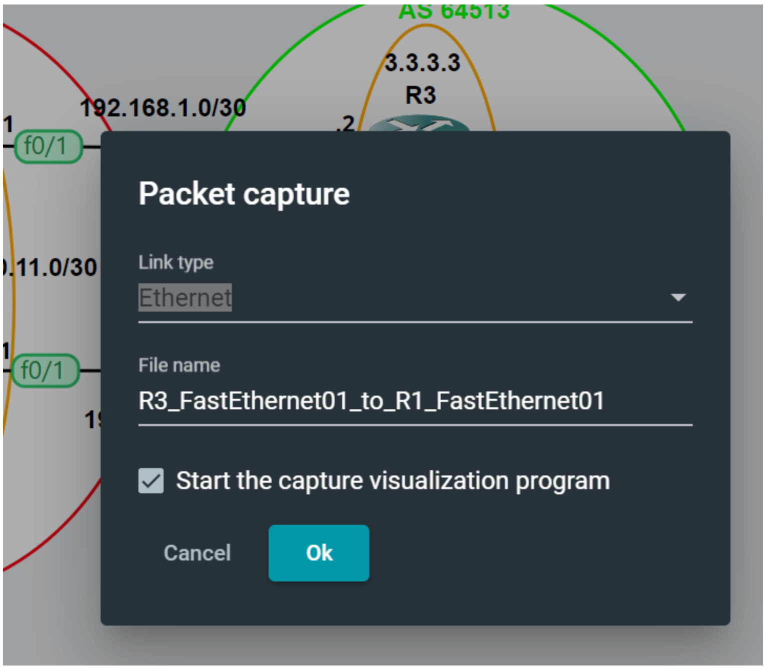
Use the commands show run | section router bgp (at the interface level), show ip bgp summary, and show ip bgp neighbors to view the configurations you have made.

Example:

```
R4#sh ip bgp summary
BGP router identifier 4.4.4.4, local AS number 64513
BGP table version is 1, main routing table version 1

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
3.3.3.3         4 64513       0       0        0    0    0 never    Active
192.168.3.1     4 64512      16      15        1    0    0 00:12:59        0
```

GNS3 also allows interconnection with Wireshark for packet capture on specific interfaces. To activate this feature, right-click and select the “start capture” option.



6 – Route advertisement via BGP

The aim is to announce the routes for devices PC1 and PC2 through their respective access routers. To do this, we will use the router bgp <ASN> command, as in point 4, but this time we will use the network <ip address> mask <mask> command to announce the terminal networks that allow connection to the two devices indicated.

R2 Configuration

```
R2 (config)# router bgp 64512
R2 (config-router)# no synchronization
R2 (config-router)# bgp log-neighbor-changes
R2 (config-router)# network 10.10.10.0 mask 255.255.255.252
```

R4 Configuration

```
R4 (config)# router bgp 64513
```

```
R4 (config-router)# no synchronization
R4 (config-router)# bgp log-neighbor-changes
R4 (config-router)# network 10.20.20.0 mask 255.255.255.252
```

7 – Checking of new routes announced via BGP

Using the show ip route bgp command on the various routers, verify that the new routes are being advertised as intended and use the show ip route bgp command to view the next hops of the advertised paths, verifying that they are as expected.

Examples:

The route 10.20.20.0 from R1 has R3 as its next hop:

```
R1#show ip route bgp
     10.0.0.0/30 is subnetted, 4 subnets
B       10.20.20.0 [20/0] via 192.168.1.2, 14:55:58
B       10.10.10.0 [200/0] via 2.2.2.2, 14:15:17
```

The route 10.20.20.0 from R2 has R4 as its next hop:

```
R2#show ip route bgp
     10.0.0.0/30 is subnetted, 4 subnets
B       10.20.20.0 [20/0] via 192.168.3.2, 00:05:00
```

```
R1#show ip bgp
BGP table version is 5, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.10.10.0/30    2.2.2.2                  0    100      0 i
*> 10.20.20.0/30    192.168.1.2                            0 64513 i
* i                 192.168.3.2              0    100      0 64513 i

R1#show ip bgp 10.20.20.0
BGP routing table entry for 10.20.20.0/30, version 3
Paths: (2 available, best #1, table Default-IP-Routing-Table)
  Advertised to update-groups:
        2
  64513
    192.168.1.2 from 192.168.1.2 (3.3.3.3)
      Origin IGP, localpref 100, valid, external, best
  64513
    192.168.3.2 (metric 20) from 2.2.2.2 (2.2.2.2)
      Origin IGP, metric 0, localpref 100, valid, internal
```

As you can see, the route from R1 to 10.20.20.0 has two paths, with the preferred one being the one that goes via R3, with the pref location denoting the default value (100).

If you view this change in Wireshark, you should identify the UPDATE message that reflects the announcement of the new routes between the BGP peers.

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
21	62.106949	c2:01:07:7c:00:01	c2:01:07:7c:00:01	LOOP	60	Reply
22	62.291648	c2:03:07:b0:00:01	c2:03:07:b0:00:01	LOOP	60	Reply
23	67.769750	192.168.1.1	192.168.1.2	BGP	100	UPDATE Message
24	67.976212	192.168.1.2	192.168.1.1	TCP	60	179 → 61001 [ACK] Seq=20 Ack=94 Win=15402 Len=0
25	72.103438	c2:01:07:7c:00:01	c2:01:07:7c:00:01	LOOP	60	Reply
26	72.298960	c2:03:07:b0:00:01	c2:03:07:b0:00:01	LOOP	60	Reply
27	82.107615	c2:01:07:7c:00:01	c2:01:07:7c:00:01	LOOP	60	Reply
28	82.294268	c2:03:07:b0:00:01	c2:03:07:b0:00:01	LOOP	60	Reply
29	92.109003	c2:01:07:7c:00:01	c2:01:07:7c:00:01	LOOP	60	Reply
30	92.297075	c2:03:07:b0:00:01	c2:03:07:b0:00:01	LOOP	60	Reply
31	93.623670	c2:01:07:7c:00:01	CDP/VTP/DTP/PAgP/UDLD	CDP	352	Device ID: R1 Port ID: FastEthernet0/1

```
> Frame 23: 100 bytes on wire (800 bits), 100 bytes captured (800 bits) on interface -, id 0
> Ethernet II, Src: c2:01:07:7c:00:01 (c2:01:07:7c:00:01), Dst: c2:03:07:b0:00:01 (c2:03:07:b0:00:01)
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.2
> Transmission Control Protocol, Src Port: 61001, Dst Port: 179, Seq: 48, Ack: 20, Len: 46
v Border Gateway Protocol - UPDATE Message
    Marker: ffffffffffffffffffffffffffffffff
    Length: 46
    Type: UPDATE Message (2)
    Withdrawn Routes Length: 0
    Total Path Attribute Length: 18
  v Path attributes
    > Path Attribute - ORIGIN: IGP
    > Path Attribute - AS_PATH: 64512
    > Path Attribute - NEXT_HOP: 192.168.1.1
  v Network Layer Reachability Information (NLRI)
    v 10.10.10.0/30
        NLRI prefix length: 30
        NLRI prefix: 10.10.10.0
```

8 – Configuring the “local preference” attribute

As mentioned in the theory sessions, BGP allows attributes to be defined for route prioritization, taking into account technical and/or economic criteria. One possibility is to create a route map in which specific attributes can be configured using the route-map command (permit default with a value of 10).

In the following configuration, the aim is to create a new route map and change the local preference parameter in R1 to a value greater than 100 (default), in order to force the next hop to reach the 10.20.20.0 network to become 192.168.1.2, which is advertised by R1 (1.1.1.1).

R1 Configuration

```
R1 (config)# route-map LOCAL_PREFERENCE_ROUTE_MAP permit 10
R1 (config-route-map)# set local-preference 200
R1 (config)# router bgp 64512
R1 (config)# neighbor 192.168.1.2 route-map LOCAL_PREFERENCE_ROUTE_MAP in
R1# clear ip bgp *
```

9 – Checking of the impact of changing the “local preference”

Use the show ip bgp command on R1 and R2 to verify that the next hop is changed to the path originating from R2, while remaining unchanged for the path originating from R1.

```
R1#sh ip bgp
BGP table version is 3, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.10.10.0/30    2.2.2.2                  0    100      0 i
*> 10.20.20.0/30    192.168.1.2                   200      0 64513 i

R1#sh ip bgp 10.20.20.0
BGP routing table entry for 10.20.20.0/30, version 3
Paths: (1 available, best #1, table Default-IP-Routing-Table)
Flag: 0x820
  Advertised to update-groups:
        2
  64513
    192.168.1.2 from 192.168.1.2 (3.3.3.3)
      Origin IGP, localpref 200, valid, external, best
```

As can be seen, the route from R2 now has R1 (1.1.1.1) as its next hop thanks to the effect of the local pref=200 parameter.

```
R2#sh ip bgp
BGP table version is 14, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.10.10.0/30    0.0.0.0                  0         32768 i
*>i10.20.20.0/30    192.168.1.2              0    200      0 64513 i
*                   192.168.3.2              0             0 64513 i
R2#sh ip bgp 10.20.20.0
BGP routing table entry for 10.20.20.0/30, version 14
Paths: (2 available, best #1, table Default-IP-Routing-Table)
Flag: 0x940
  Advertised to update-groups:
        1
  64513, (received & used)
    192.168.1.2 (metric 20) from 1.1.1.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 200, valid, internal, best
  64513
    192.168.3.2 from 192.168.3.2 (4.4.4.4)
      Origin IGP, metric 0, localpref 100, valid, external
```

10 – Configuration of the "multi exit discriminator (med)" attribute

In general, BGP offers more options for configuring outgoing traffic than incoming traffic. One way to influence the traffic coming in from a neighboring AS is to use the multi discriminator exit (med) attribute.

In this case, the intention is for AS 64512 to try to influence AS64513 to choose the peering relationship between R1<->R3 to reach PC1. However, as can be seen by using the show ip bgp 10.10.10.0 command on R4, the default route for entry into AS 64512 is through the R2<->R4 peering relationship, contrary to what we want to implement.

```
R4#show ip bgp 10.10.10.0
BGP routing table entry for 10.10.10.0/30, version 9
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to update-groups:
        2
  64512
    192.168.1.1 (metric 20) from 3.3.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, internal
  64512
    192.168.3.1 from 192.168.3.1 (2.2.2.2)
      Origin IGP, metric 0, localpref 100, valid, external, best
```

As can be seen, router R2 is the path preferably selected by BGP to forward traffic from AS 64513 to route 10.10.10.0. Therefore, in order for this routing to be done preferably through R1, it will be necessary to implement a new route map on this router that changes the current attributes of the "MED" attribute, bearing in mind that the route with the lowest MED metric value is selected with the highest priority. Thus, a higher metric is introduced in R2 to prioritize R1.

R2 Configuration

```
R2 (config)# route-map MED_ROUTE_MAP permit 10
R2 (config-route-map)# set metric 111
R2 (config)# router bgp 64512
R2 (config)# neighbor 192.168.3.2 route-map MED_ROUTE_MAP out
R2# clear ip bgp *
```

11 – Checking of the impact of the change in the "multi exit discriminator (med)" attribute

By performing a new check using the commands show ip route 10.10.10.0 and show ip bgp 10.10.10.0 on R4, it is possible to verify that the entry point into AS 64512 is made by interface 192.168.1.1 on R1, originating from R3 (3.3.3.3).

It is also clear that the route metric via R2 (2.2.2.2) is ignored because it has a higher metric (111).

```
R4#sh ip route 10.10.10.0
Routing entry for 10.10.10.0/30
  Known via "bgp 64513", distance 200, metric 0
  Tag 64512, type internal
  Last update from 192.168.1.1 00:01:34 ago
  Routing Descriptor Blocks:
  * 192.168.1.1, from 3.3.3.3, 00:01:34 ago
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 64512

R4#show ip bgp 10.10.10.0
BGP routing table entry for 10.10.10.0/30, version 13
Paths: (2 available, best #1, table Default-IP-Routing-Table)
Flag: 0x4960
  Advertised to update-groups:
        1
  64512
    192.168.1.1 (metric 20) from 3.3.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, internal, best
  64512
    192.168.3.1 from 192.168.3.1 (2.2.2.2)
      Origin IGP, metric 111, localpref 100, valid, external
```